

Ziwei Wang

Email: vivi@hust.edu.cn | [Google Scholar](#) | [Homepage](#)

Ph.D. Candidate, School of Artificial Intelligence and Automation, Huazhong University of Science and Technology

RESEARCH INTERESTS

Deep learning; Brain-computer interfaces; Rehabilitation and healthcare foundation models

EDUCATION

Sep. 2021-Present	Huazhong University of Science and Technology	Intelligent Science and Technology	Ph.D. Candidate
Sep. 2017-Jun. 2021	Central South University	Measurement and Control Technology and Instrumentation	B.Eng.

Advisor: Prof. Dongrui Wu, NSFC Distinguished Young Scholars Fund; IEEE Fellow; Editor-in-Chief, IEEE TFS (IF=11.9)

SELECTED HONORS AND AWARDS

- 2025 Ant Group InTech Scholarship (10 awardees worldwide; 2 in Digital Medicine)
- 2025 National Scholarship (Ph.D.)
- 2024 National Scholarship (Ph.D.)
- 2021 National Scholarship (Undergraduate)
- 2025 Merit Student Pacesetter (Highest student honor at HUST)
- 2022 IEEE Computational Intelligence Society Student Grant (5 awardees worldwide)
- 2025 Top 10 Advances in Brain-Computer Interfaces in China (Huanau Award)
- 2024 Top 10 Highlights in Brain-Computer Interfaces in China (Huanau Award)
- 2023 National First Prize in World Robot Contest
- 2025/2021 National Second Prize in World Robot Contest
- 2021 Outstanding Graduate of Hunan Province
- 2024 IOP Trusted Reviewer

PUBLICATIONS

First-author publications

- [1] Z. Wang, et al., "Canine EEG helps human: Cross-species and cross-modality epileptic seizure detection via multi-space alignment," National Science Review, vol. 12, no. 6, p. nwaf086, 2025. (First author; CAS Q1 Top; IF=17.1)
- [2] Z. Wang, et al., "DBConformer: Dual-branch convolutional Transformer for EEG decoding," IEEE Journal of Biomedical and Health Informatics, vol. 30, no. 5, pp. 4134-4147, 2026. (First author; JCR Q1; CAS Q1 Top; IF=6.8)
- [3] Z. Wang, et al., "CKD: Contrastive knowledge distillation for cross-dataset EEG classification," IEEE Transactions on Biomedical Engineering, Early Access, pp. 1-13, doi: 10.1109/TBME.2026.3701548, 2026. (First author; CAA-A; IF=4.5)
- [4] Z. Wang, et al., "Time-frequency transform based EEG data augmentation for brain-computer interfaces," Knowledge-Based Systems, vol. 311, p. 113074, 2025. (First author; CAS Q1 Top; IF=7.6)
- [5] Z. Wang, et al., "MVCNet: Multi-view contrastive network for motor imagery classification," Knowledge-Based Systems, vol. 328, p. 114205, 2025. (First author; CAS Q1 Top; IF=7.6)
- [6] Z. Wang†, S. Li†, et al., "Channel reflection: Knowledge-driven data augmentation for EEG-based brain-computer interfaces," Neural Networks, vol. 176, p. 106351, 2024. (Co-first author; CAS Q1 Top; IF=6.3)
- [7] Z. Wang, et al., "Unsupervised domain adaptation for cross-patient seizure classification," Journal of Neural Engineering, vol. 20, no. 6, p. 066002, 2023. (First author; CAA-A; IF=4.5)
- [8] Z. Wang, et al., "Synthetic data generation for brain-computer interfaces: Overview, benchmarking, and future directions," Artificial Intelligence Review, Under review, 2026. (First author; CAS Q1 Top)
- [9] Z. Wang†, X. He†, T. Jia, H. Wang, and D. Wu*, "FAConformer: Frequency-aware convolutional Transformer for auditory attention decoding," Science China Information Sciences, Under review, 2026. (Co-first author; CAS Q1 Top)
- [10] Z. He†, Z. Wang†, and D. Wu*, "DenoNet: A GAN-VAE hybrid generative model for SSVEP denoising," Engineering, Under review, 2026. (Co-first author; CAS Q1 Top)
- [11] Z. Wang, S. Li, and D. Wu*, "Cross-species epileptic seizure detection via multi-space alignment," China Systems Science Congress, Oral, Kunming, China, May 2025. (First author; CAA-C)
- [12] Z. Wang and D. Wu*, "EEG-based seizure prediction using transfer learning," IEEE Computational Intelligence Society, 2021. (First author; Student Grant Paper)

Other selected publications

- [1] ..., Z. Wang, ..., "PGAP: Purity-guided active prompting for EEG decoding with LLMs," IEEE Computational Intelligence Magazine, Early Access, 2026. (Second author; JCR Q1; IF=11.2)
- [2] ..., Z. Wang, ..., "Black-box test-time ensemble," IEEE Computational Intelligence Magazine, Early Access, 2025. (Second author; JCR Q1; IF=11.2)
- [3] ..., Z. Wang, ..., "Alignment-based adversarial training (ABAT) for improving the robustness and accuracy of EEG-based BCIs," IEEE Trans. on Neural Systems and Rehabilitation Engineering, vol. 32, pp. 1703-1714, 2024. (Second author; JCR Q1)
- [4] ..., Z. Wang, ..., "T-TIME: Test time information maximization ensemble for plug-and-play BCIs," IEEE Trans. on Biomedical Engineering, vol. 71, pp. 423-432, 2023. (Second author; CAA-A)
- [5] ..., Z. Wang, ..., "Multi-source decentralized transfer for privacy-preserving BCIs," IEEE Trans. on Neural Systems and Rehabilitation Engineering, vol. 30, pp. 2710-2720, 2022. (Second author; JCR Q1)
- [6] ..., Z. Wang, ..., "Deep source semi-supervised transfer learning (DS3TL) for cross-subject EEG classification," IEEE Trans. on Biomedical Engineering, vol. 71, pp. 1308-1318, 2023. (Third author; CAA-A)

- [7] ..., **Z. Wang**, ..., "SDDA: Spatial distillation based distribution alignment for cross-headset EEG classification," IEEE Trans. on Biomedical Engineering, Early Access, 2026. (Third author; CAA-A)
- [8] ..., **Z. Wang**, ..., "SACM: SEEG-audio contrastive matching for Chinese speech decoding," IEEE Trans. on Biomedical Engineering, Early Access, doi: 10.1109/TBME.2026.3678858, 2026. (Fourth author; CAA-A)
- [9] ..., **Z. Wang**, ..., "User-wise perturbations for user identity protection in EEG-based BCIs," Journal of Neural Engineering, vol. 22, no. 1, p. 016040, 2025. (Fourth author; CAA-A)
- [10] ..., **Z. Wang**, ..., "Discriminative prototype learning (DPL) for cross-subject EEG classification," IEEE Trans. on Pattern Analysis and Machine Intelligence, Reject and resubmit, 2026. (Second author; CAS Q1 Top)
- [11] ..., **Z. Wang**, ..., "SplitFL: Split federated learning for cross-silo EEG decoding," Information Fusion, Under review, 2026. (Third author; CAS Q1 Top)
- [12] ..., **Z. Wang**, ..., "Backpropagation-free test-time adaptation for lightweight brain-computer interfaces," IEEE Journal of Biomedical and Health Informatics, Under review, 2026. (Fourth author; CAS Q1 Top)
- [13] ..., **Z. Wang**, ..., "Effective and efficient intracortical brain signal decoding with spiking neural networks," Neurocomputing, Under review, 2026. (Fifth author; CAS Q1 Top)
- [14] ..., **Z. Wang**, ..., "RAICL: Retrieval-augmented in-context learning for vision-language-model based EEG seizure detection," IEEE Systems, Man, and Cybernetics, Under review, 2026. (Fifth author; CAS Q1 Top)

CONFERENCE PRESENTATIONS

- [1] **Z. Wang** et al., "Transfer Learning for EEG-based Brain-Computer Interfaces: Tutorial," ICONIP 2023, Tutorial, Changsha, China, Nov. 2023.
- [2] **Z. Wang** et al., "EEG-based automatic seizure detection," Alibaba Yunqi Conference 2024, AI for Science Industry-University-Research Innovation Showcase, Oral, Hangzhou, China, Sep. 2024.
- [3] **Z. Wang** et al., "Cross-species and cross-modality epileptic seizure detection via multi-space alignment," China Systems Science Congress, Oral, Kunming, China, May 2025.
- [4] **Z. Wang** et al., "Channel Reflection: Knowledge-Driven Data Augmentation for EEG-Based BCIs," ChinaBMI 2024, Poster, Hangzhou, China, Dec. 2024.
- [5] **Z. Wang** et al., "Channel Reflection: Knowledge-Driven Data Augmentation for EEG-Based BCIs," SAAC 2024, Poster, Wuhan, China, Dec. 2024.
- [6] **Z. Wang** et al., "DBConformer: Dual-branch convolutional Transformer for EEG decoding," SAAC 2025, Poster, Wuhan, China, Dec. 2025.

RESEARCH GRANTS

National Natural Science Foundation of China, Young Student Basic Research Program (Ph.D. Track), Grant No. 625B2077, "Cross-Species Multimodal EEG General Representation and Foundation Model", 300,000 RMB Jan. 2026-Dec. 2027

China Association for Science and Technology, Young Elite Scientists Sponsorship Program - Doctoral Student Special Plan, "EEG General Foundation Model", 40,000 RMB Jan. 2026-Dec. 2027

INTERNSHIP EXPERIENCE

Alibaba Cloud Group Oct. 2022-Apr. 2023

- Proposed a Sparse Attention module to improve modeling efficiency and predictive performance for long-sequence forecasting.
- Designed a domain-generalized mixture-of-experts model to improve robust fault prediction under temporal distribution shifts and device-level heterogeneity.
- Designed five N-gram and TF-IDF based data augmentation operators for anomaly-aware data augmentation.

ACADEMIC SERVICE

Reviewer for IEEE Transactions on Fuzzy Systems, IEEE Journal of Biomedical and Health Informatics, IEEE Transactions on Neural Systems and Rehabilitation Engineering, IEEE Transactions on Biometrics, Behavior, and Identity Science, Knowledge-Based Systems, Journal of Neural Engineering, Scientific Reports, Cognitive Computation, ICONIP, and IEEE SMC; recognized as an IOP Trusted Reviewer.

PATENTS

- [1] D. Wu and **Z. Wang**, "A cross-subject target-domain proxy subdomain adaptation method, system, and medium," Chinese Invention Patent, Publication No. CN115590530A, granted. (First student contributor)
- [2] D. Wu, **Z. Wang**, S. Li, and X. Chen, "A supervised contrastive learning method and device based on spatio-temporal-frequency consistency," Chinese Invention Patent, Publication No. CN118626901A, published. (First student contributor)
- [3] D. Wu and **Z. Wang**, "A model training method for fused contrastive knowledge distillation in cross-dataset motor imagery EEG decoding," Chinese Invention Patent, Application No. AJ2601344. (First student contributor)
- [4] D. Wu, **Z. Wang**, and Z. He, "A denoising model training method, denoising method, and data augmentation method for EEG signals," Chinese Invention Patent, Application No. 202610308697.1. (First student contributor)
- [5] D. Wu, S. Li, **Z. Wang**, and S. Zheng, "A construction method, EEG classification method, and system for cross-device EEG classification models," Chinese Invention Patent, Publication No. CN119719914A, granted. (Second student contributor)
- [6] D. Wu, S. Li, and **Z. Wang**, "A test-time ensemble method, medium, and device for black-box models," Chinese Invention Patent, Publication No. CN118626848A, published. (Second student contributor)
- [7] D. Wu, D. Liu, S. Li, and **Z. Wang**, "A construction method, EEG signal classification method, and system for cross-headset EEG classification models," Chinese Invention Patent, Publication No. CN120067794A, published. (Third student contributor)